

Chapter 2 Essential Principles of Signals on Interconnects

With just a few essential principles based on Maxwell's equations, we can gain the insight and intuition needed to understand how signals interact with interconnects. This will enable us to understand how physical design affects the important electrical properties of the interconnects that influence how signals are treated.

After all, the design process is a creative process. While we will always leverage analytical methods such as approximations and numerical simulations, we start the design process using our engineering judgement and intuition.

The better calibrated our engineering intuition, the farther along a successful design strategy we can start each project.

2.1 All Interconnects are Transmission Lines

In Chapter 1 we introduced the concept of a transmission line. Without exception, all interconnects used in any electronic system to connect a signal between two nodes or terminals is a transmission line.

Every transmission line always has two conductors. We label one of them a signal path and the other the return path.

Don't get in the habit of calling the return path ground. The ground path or a ground plane may be used as the return path, but this is a special case. Any conductor with any DC voltage may be used as a return path.

Often, the return path is called the *reference conductor*. This is very misleading and confusing. What does this mean? Can I

identify any conductor as the reference? How do I tell the signal which conductor I selected as the reference?

The term *reference conductor* is also ambiguous. The words we use matter. They affect our intuition and how we conceptualize the conductors.

Get in the habit of calling the other conductor in a transmission line the return path and you will train your intuition to think about the return currents being in this conductor.

As we will show in this chapter, the selection and design of the return path affects the signal propagation just as much as the signal path. If you only engineer the signal path, and do not engineer the return path, the signal will absolutely find a return path on its own.

Chances are, you will not like the return path the signal finds for itself. The signal propagation will not be as clean and low noise as if you engineered the return path on purpose.

2.2 Signals Are Dynamic

The purpose of an interconnect is to transport a signal from a transmitter to a receiver with acceptable distortion. While we focus on the design and characterization of the interconnects in this book, it is the quality of the signal that is transported by the interconnect that motivates our design choices.

When we launch a voltage between the signal and return path at the beginning of the transmission line, we are changing the electric field between these two conductors at the front end. A changing electric field generates a changing magnetic field, which then generates a changing electric field.

Once begun, these fields are *self-propagating*. They will propagate down the transmission line between the signal and return path conductors passing through the dielectric between them. This property of self-propagating changing electric fields is literally built into the fabric of space-time.

It was one of the profound contributions of James Clerk Maxwell to realize that changing electric and magnetic fields is really the nature of light. The propagating signal on a transmission line is literally light, though typically at a frequency in the radio spectrum of 1 MHz to 50 GHz, rather than the 500,000 GHz of red light.

It is this fundamental property of electromagnetic fields that creates the signal that propagates down the transmission line.

The signal that propagates is the changing electric and magnetic fields associating with the changing voltage between the signal and return path. The speed of the signal is really about how quickly the changing electric field between the signal and return path can create the changing magnetic field, which then generates the changing electric field in the dielectric material and self-propagates down the transmission line. This is illustrated in **Figure 2.1**.

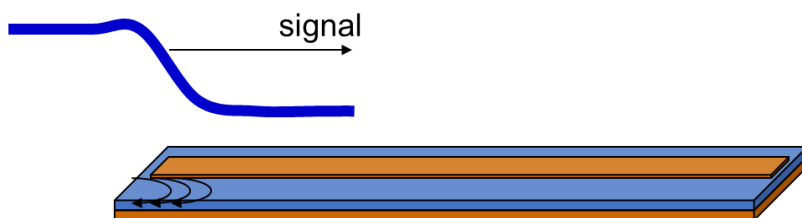


Figure 2.1 The changing and propagating electric fields between the signal and return path and the resulting instantaneous voltage profile that would be measured between the signal and return conductors.

As soon as the voltage is applied and ramps up between the signal and return conductors at one end of the transmission line, this changing voltage will create a changing electric field that will

propagate down the line. This property is built into the very fabric of space-time.

There is absolutely nothing you can do to prevent this applied, changing voltage from propagating down the transmission line between the signal and return path at the speed of light in the material.

It is often incorrectly believed that the signal is traveling in the copper traces and is related to the resistance of the traces. Using this model, you would be inclined to believe that the speed of a signal is related to the resistance of the traces. A lower resistance would result in a faster speed. This is totally incorrect.

The signal is really the propagating, changing electric field between the signal and return path. This appears as a voltage between the signal and return path. This is why it's the dielectric material between the signal and return path that affects the speed of the signal.

In air, the speed of a changing, self-propagating electric field is 3×10^8 m/sec or 30 cm/nsec or 11.8 inches/nsec. If the electric field is in a dielectric material, the dielectric constant of the material will slow the speed of propagation of the changing electric by the square root of the dielectric constant.

The speed of the signal (light) in a dielectric is:

$$v = \frac{11.8 \text{ in/nsec}}{\sqrt{Dk}} = \frac{30 \text{ cm/nsec}}{\sqrt{Dk}} \quad (2.1)$$

where

11.8 in/nsec is the speed of light in air

Dk = the dielectric constant of the laminate material.

With a typical dielectric constant for most laminates around 4, the typical speed of a signal is about 6 inches/nsec or 15 cm/nsec.

This is a very important rule of thumb to remember. The speed of a signal propagating down a transmission line made from an FR4 laminate is about 6 inches per nsec.

A signal launched into a 12-inch-long transmission line on a circuit board would take about $12 \text{ inches} / 6 \text{ inches/nsec} = 2 \text{ nsec}$ to enter one end and reach the other end.

After the signal is launched into the transmission line, this voltage difference will propagate down the interconnect at the speed of light in the material.

If we could freeze time, we would see the voltage distribution across the transmission line such as shown in **Figure 2.2**.

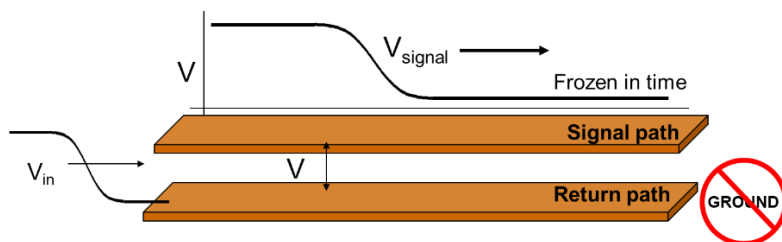


Figure 2.2 The voltage distribution on a transmission line frozen in time. The signal is this changing voltage that is propagating.

The signal is the voltage transition that is propagating down the transmission line with its associated changing electric field at the leading edge. The signal is, by its nature, dynamic, constantly in motion, propagating down the transmission line.

It is important to make the distinction between the voltage that would be measured on the transmission line between the two conductors, with an oscilloscope, for example, and the signal that propagates.

At any instant, there is a voltage difference between every two adjacent points on the signal and return path. If we were to place a scope probe between the signal and return path, the scope would measure the actual, instantaneous voltage between these two points.

However, the measurement of the voltage between two adjacent points alone tells us nothing about the *direction of propagation* of the signal. We sometimes refer to this voltage measured by a scope as a *scalar* voltage in that it is just one number, with no direction. This is distinct from the description of a *vector*, which would have direction consisting of propagation information.

We call the voltage pattern that is propagating on the interconnect, with information about its direction, the signal. It is the signal that is the propagating voltage. The voltage itself is just the scalar value as would be measured with a scope.

When there is just one signal propagating on the line, the voltage would be the same magnitude as the signal. But, when multiple signals propagate on the line. However, in opposite directions, each signal is clear and distinct, but a voltage measurement cannot distinguish between them. The voltage measured by the scope would be the sum of the propagating signals.

A scope measurement of the voltage on a transmission line tells us nothing about the direction of propagation of the signal. The propagating nature of the signal is an important feature of the signal and how it interacts with the interconnect, yet a scope measurement tells us nothing about this property.

If we learn all of our signal integrity just from scope measurements, we will have missed one of

the most important properties of signals and our intuition may be completely wrong.

2.3 A Simple Free Tool to Illustrate the Propagation of a Signal

Signals are dynamic. It is hard to visualize this behavior on a flat, static page. To help calibrate your intuition, you can [download this free simulation tool](#) to show the dynamic nature of a signal.

[Watch this video and I will walk you through using this simulation tool.](#)

This simple simulation tool, based on flash and written by Yoshi Tsuji of Teledyne LeCroy, will be used throughout this book to illustrate some of the dynamic properties of signals propagating on transmission lines. While it has a simulation engine at its core, it is not a substitute for a circuit simulator like a SPICE-based simulator. It is really designed to illustrate the general principles of signals propagating on a transmission line and interacting with the instantaneous impedance it encounters.

In this first example we will use it simply to show the voltage on a transmission line as it propagates. **Figure 2.3** is a screenshot showing two instances of the same signal on a transmission line.

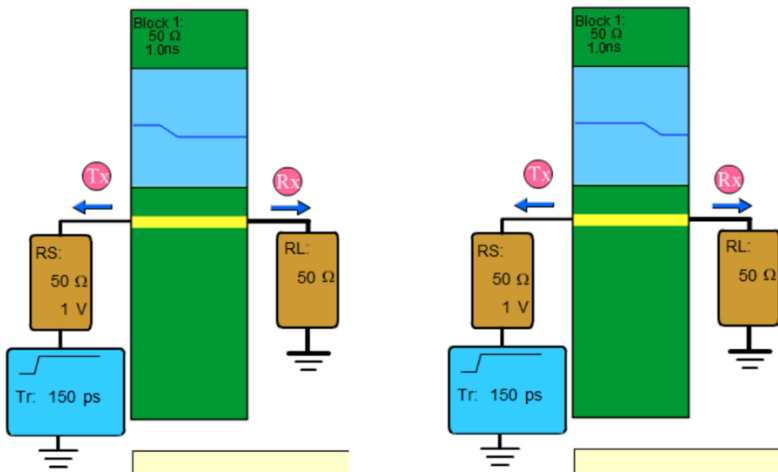


Figure 2.3 Two snapshots of a signal propagating down a transmission line using the free downloaded simulator.

The green region represents the return path of the transmission line. The yellow strip is the top view of the microstrip transmission line. It is driven with an adjustable ramp source with a source impedance. It is connected to an adjustable termination impedance.

Once the signal is launched from the source, the dark blue trace in the light blue field is a plot of the voltage profile along the microstrip transmission line as short time intervals as the signal propagates.

This tool allows you to visualize the rising edge of the signal as it propagates down this transmission line. Later in this book we will use this tool to visualize the reflections of the signal as it encounters changes in the instantaneous impedance.

2.4 Time Delay and Wiring Delay

An important property of a transmission line is its *time delay*. This is how long it takes a signal to enter the line and come out the other end. The time delay, also referred to as the *propagation delay*, of a transmission line is related to the length of the line and the speed of the signal by:

$$TD[n\text{ sec}] = \frac{\text{Len}[\text{in}]}{v[\text{in} / n\text{ sec}]} \quad (2.2)$$

where

Len[in] = the physical length of the transmission line in inches

v[in/nsec] = the speed of the signal in the dielectric of the transmission line, roughly 6 inches/nsec in FR4 materials.

We sometimes refer to the *wiring delay* of an interconnect as $1/v =$ roughly $1/6[\text{in}/\text{nsec}] = 0.17 \text{ nsec}/\text{inch}$, in FR4. This is the delay per inch of the interconnect. It is not a speed, but a delay per inch, the inverse of the speed.

The time delay of a transmission line in FR4, using the wiring delay, is

$$TD[n\text{ sec}] = \text{Len}[\text{in}] \times 0.17[n\text{ sec}/\text{inch}]$$

For example, an interconnect 2 inches long will have a delay of $2\text{-inches} \times 0.17 \text{ nsec}/\text{inch} = 0.34 \text{ nsec}$.

Both the speed of a signal as 6 inches/nsec and the wiring delay of 0.17 nsec/inch are important rules of thumb to remember. They will help us quickly estimate the time delay of an interconnect.

2.5 Signals See an Instantaneous Impedance

The most important electrical property of an interconnect that determines the relationship between the voltage and current on it is its impedance. What makes this term so confusing is that there are multiple types of impedance, each with a different definition and each with a different application.

Regardless of the type of impedance, all definitions have their origin in the simple definition,

$$Z = \frac{V}{I} \quad (2.3)$$

As the signal propagates on a transmission line, it sees an *instantaneous* impedance each step along the way. We can easily estimate this instantaneous impedance using a simple *physics model* for the transmission line and this definition of impedance.

The V term in the definition is just the signal voltage on the transmission line, 1V, for example. To determine the impedance the signal sees, we need to find the current associated with the signal as it travels down the transmission line.

The simplest path to determine the instantaneous impedance the signal sees is to take a zen approach and *be the signal* to see what it sees as it propagates.

[Watch this video and I walk you through the analysis of the instantaneous impedance and the characteristic impedance of a transmission line.](#)

Imagine, as in **Figure 2.4**, that we are the signal walking on the transmission line. Ahead of us, there is no voltage on the transmission line. Behind us, there is a 1V voltage we left in our wake between the signal and return path.

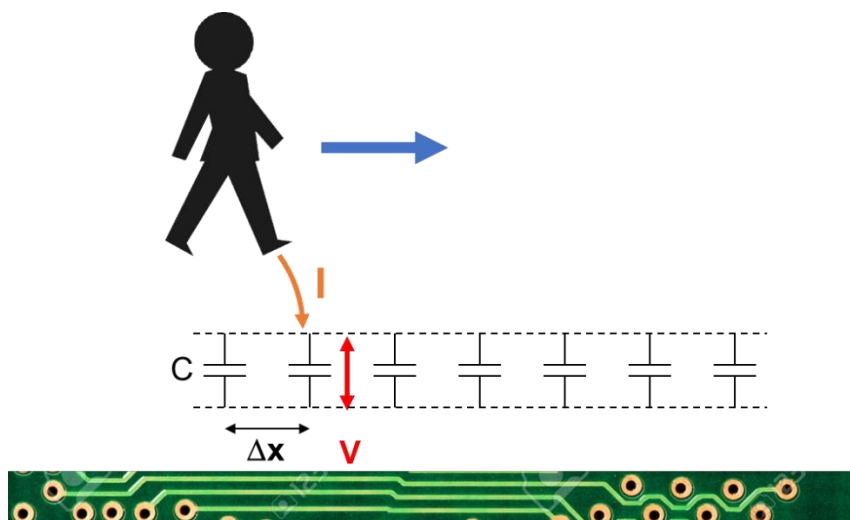


Figure 2.4 Imagine you are the signal, walking down the transmission line charging up the line as you go. Each step has a capacitance between the signal and return path.

In each footstep, there is some capacitance between the signal and return path. If the interconnect is uniform, and we travel down the interconnect at the same speed, the length of our footstep is the same and the capacitance in each footstep is the same. The capacitance in each footstep is

$$C = C_{\text{Len}} \times \Delta x = C_{\text{Len}} \times \Delta t \times v \quad (2.4)$$

where:

C = the capacitance in each footstep

C_{Len} = the capacitance per length of the transmission line

Δx = the length of each footstep

Δt = the time it takes to step from footstep to footstep

v = the speed of the signal- the speed of light in the dielectric

With each step we take, we are charging up this capacitance in each footstep from 0V to 1V. There is some amount of charge, Q , we need to dump into the capacitor to raise it to 1V.

$$\Delta Q = C \times V \quad (2.5)$$

As we step along the line, depositing the same charge in each footstep to leave the 1V in our wake, taking the same time between footsteps, there is a current coming out of our foot equal to:

$$I = \frac{\Delta Q}{\Delta t} = \frac{C \times V}{\Delta t} = \frac{C_{\text{Len}} \times \Delta t \times v \times V}{\Delta t} = C_{\text{Len}} \times v \times V \quad (2.6)$$

As long as the transmission line is uniform, this current coming out of our foot to charge up the transmission line is constant.

This is a profound observation. As the signal travels down the uniform transmission line, there is a constant current associated with the signal that successively charges up each section of the transmission line in order to leave a 1V signal in its wake. This makes the instantaneous impedance the signal sees constant.

The ratio of the constant voltage of the signal to the constant current associated with it is the instantaneous impedance the signal sees. This instantaneous impedance is

$$Z = \frac{V}{I} = \frac{V}{C_{\text{Len}} \times v \times V} = \frac{1}{C_{\text{Len}} \times v} \quad (2.7)$$

As we walk down the line, if the cross section is uniform, the capacitance per length and the speed of the signal is constant so the instantaneous impedance is constant. The signal sees one value of instantaneous impedance as it propagates.

If we double the length of the transmission line, the instantaneous impedance will not change. It will take us longer to travel down the transmission line, but each step will see the same instantaneous impedance.

If the line width were to increase in one region of the transmission line, the capacitance of that footstep would increase. The charge required to increase this capacitance to the same 1V would increase. If we take the same time to walk from footstep to footstep, and the charge has increased, the current required to charge up each footstep will increase. If the current increases for the same voltage signal, the instantaneous impedance will have decreased.

***As the signal propagates down the uniform transmission line, if the capacitance per length increases, the instantaneous impedance decreases.
If the capacitance per length decreases, the instantaneous impedance increases.***

We all have intuition about how the physical design of an interconnect affects the capacitance per length. A larger width means a higher capacitance per length. The thicker the dielectric

spacing between the signal and return, the lower the capacitance per length.

We can develop the same physical intuition about the impact from geometry and material properties on the instantaneous impedance by just inverting the behavior. Whatever feature increases the capacitance per length will decrease the instantaneous impedance. Whatever feature decreases the capacitance per length increases the instantaneous impedance.

2.6 Instantaneous Impedance and Characteristic Impedance

A signal sees an instantaneous impedance of a transmission line as it propagates down a transmission line. If the transmission line is uniform, the instantaneous impedance is constant down the length. There is one value of instantaneous impedance that characterizes a uniform transmission line.

We call this one value of instantaneous impedance the *characteristic impedance* of the uniform transmission line.

There are two figures of merit that describe any uniform transmission line:

- The time delay
- The characteristic impedance

These are the two figures of merit of an ideal, lossless, uniform transmission line and is the basis of the T-element model common to all versions of SPICE.

If we know these two properties of a transmission line, we know everything important about it and can predict how a signal will propagate on the transmission line.

The characteristic impedance, like the instantaneous impedance, does not depend on the length of the transmission line. It is only about the cross section and material properties.

The time delay of the transmission line will depend on the length of the transmission line.

We translate this cross-section information and material properties into the characteristic impedance using analysis tools. The most important analysis tool is a 2-D field solver.

This sort of tool will take the cross-section geometry information and calculate the electric field distribution given the boundary conditions of the conductors and dielectrics. It literally solves Maxwell's equations to calculate the electric field distribution and from this, the instantaneous impedance that is the characteristic impedance.

If length is provided, it will also calculate the time delay of the transmission line.

Figure 2.5 is an example of the input and output of the Polar Instruments SI9000, a popular 2-D field solver, for the special geometry of an uncoated microstrip transmission line. With a line width of 10 mils and a dielectric thickness of 5 mils, a 2:1 aspect ratio, the characteristic impedance of this microstrip is 47 ohms.

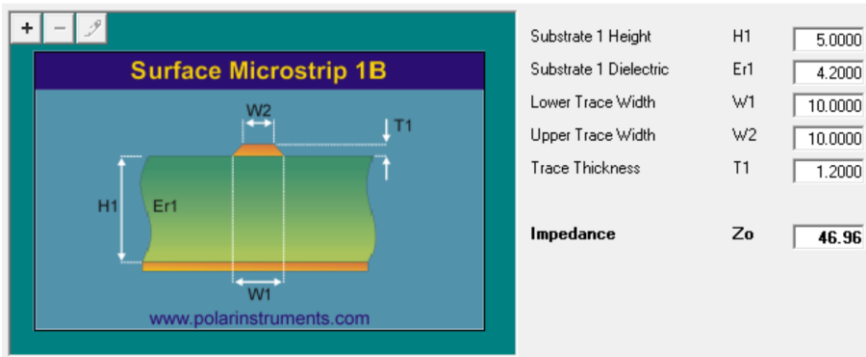
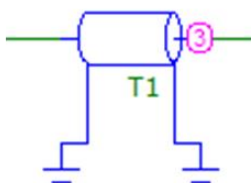
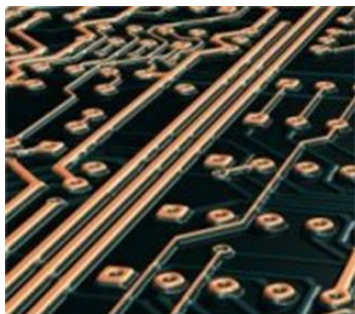


Figure 2.5 An example of the calculated characteristic impedance from geometry input information from Polar Instruments SI9000.

Every uniform transmission line has a characteristic impedance and time delay. Whether the transmission line is a 10-km long coax cable used to carry cable TV signals, or a 1-inch trace on a circuit board, each uniform interconnect has one value of characteristic impedance. This is illustrated in **Figure 2.6**.



An ideal
transmission line
model: Z_0 , TD

Figure 2.6 A uniform transmission line can be an inch long or a km long and be described with the same two figures of merit, characteristic impedance, and time delay.

If the interconnect is not uniform, the instantaneous impedance will vary down its length and there is no one value of instantaneous impedance that characterizes it. It has no characteristic impedance.

By definition, a characteristic impedance ONLY applies to a uniform transmission line. If the interconnect is not uniform, we may separate it into uniform sections and define the characteristic impedance of each uniform section.

If the instantaneous impedance varies a little, we can still approximate the transmission line as a uniform transmission line with a characteristic impedance within some limit. **Figure 2.7**

shows an example of the measured TDR response of five different RG174 coax cables. This is a direct measure of the instantaneous impedance of the transmission lines, showing the instantaneous impedance varying between the cables and constant to about $\pm 1\text{ohm}$ down their length. Each cable can be described by a characteristic impedance and time delay.

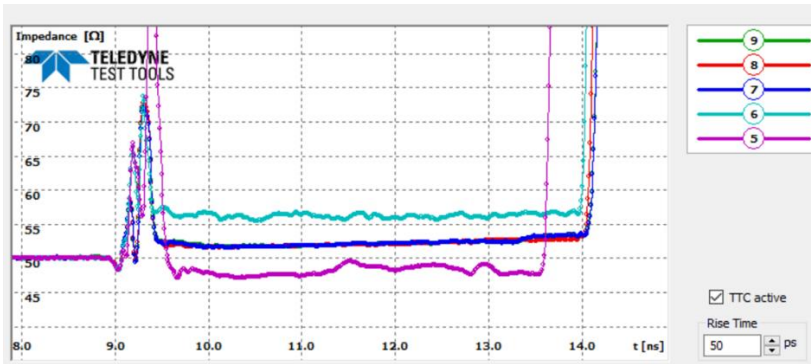


Figure 2.7 Example of the measured instantaneous impedance of five different RG174 coax cables.

2.7 What Happened to the Inductance of the Transmission Line?

In this analysis of the instantaneous impedance and the characteristic impedance of a transmission line, there was no mention of any *inductance* associated with the transmission line. We introduced the concept of the capacitance per length of the transmission line and the speed of propagation of the signal on the transmission line.

Where does the inductance of the transmission line come in?

When the electrical properties of a physical transmission line are described in other textbooks, they are often introduced in terms of the lumped circuit (LC) elements. This is often referred to as the *n-section lumped circuit model*. An example is shown in **Figure 2.8**.

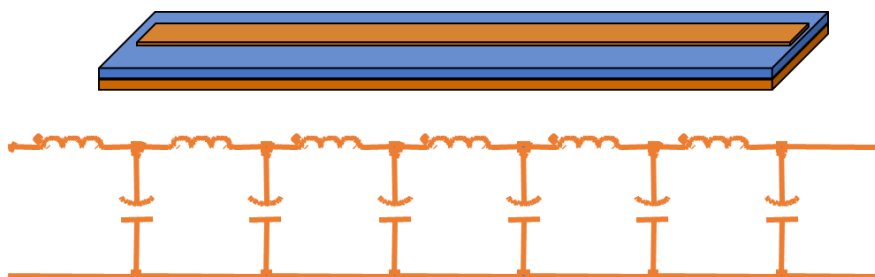


Figure 2.8 A common electrical model (bottom) for a physical transmission line (top) based on a lumped circuit LC ladder model.

The physical transmission line is approximated as a combination of n -capacitor, C , elements and n -inductor, L , circuit elements. This equivalent electrical circuit model is an approximation to the actual electrical behavior of a real physical transmission line. It can be a good approximation, in some situations, if used correctly. It is equally likely to be a poor approximation in other situations.

As is pointed out in Chapter 8, this is an approximation to a uniform transmission line, but is not a very good approximation *compared to* the ideal T-element model of a transmission line, which every SPICE simulator understands. The T-element circuit model is based on the physics model. This n -section lumped circuit model is introduced here only to clarify the role of the inductance associated with the transmission line.

In this model, there are three different inductance-related elements with the following parameters:

L_{total} = the total loop inductance between the signal and the return path

L_{len} = the length of the transmission line

L_{Len} = the inductance per length in the transmission line

ΔLen = the small length of the transmission line to which the inductance elements in the model correspond

L = the small inductance of each element used in the equivalent circuit model

n = the number of sections into which the transmission line is broken

The connection between all of these is:

$$\text{Len} = \Delta\text{Len} \times n$$

$$L_{\text{total}} = L \times n \quad (2.8)$$

$$L_{\text{Len}} = \frac{L_{\text{total}}}{\text{Len}}$$

There are comparable sets of capacitance elements:

C_{total} = the total capacitance between the signal and the return path

Len = the length of the transmission line

C_{Len} = the capacitance per length in the transmission line

ΔLen = the small length of the transmission line for which the inductance elements in the model correspond

C = the small capacitance in each circuit element used in the equivalent circuit model

n = the number of sections into which the transmission line is broken

The connection between all of these is:

$$L_{\text{len}} = \Delta L_{\text{len}} \times n$$

$$C_{\text{total}} = C \times n \quad (2.9)$$

$$C_{\text{Len}} = \frac{C_{\text{total}}}{L_{\text{len}}}$$

Using the equivalent lumped circuit model, we can derive a time delay of the transmission line, the speed of propagation, and a characteristic impedance. The process is to rewrite each circuit element with its connection between the voltage across it and the current through it:

$$\frac{\partial}{\partial x} V(x, t) = -L \frac{\partial}{\partial t} I(x, t)$$

and (2.10)

$$\frac{\partial}{\partial x} I(x, t) = -C \frac{\partial}{\partial t} V(x, t)$$

These coupled, first-order differential equations are combined to create a linear second-order differential equation, the telegrapher's equation, for the voltage and current distributions on the transmission line:

$$\frac{\partial^2}{\partial t^2} V(x, t) = \frac{1}{LC} \frac{\partial^2}{\partial x^2} V(x, t)$$

and (2.11)

$$\frac{\partial^2}{\partial t^2} I(x, t) = \frac{1}{LC} \frac{\partial^2}{\partial x^2} I(x, t)$$

The telegrapher's second-order, linear differential equation is solved with one solution being sine waves of current and voltage.

The mathematical details of deriving and solving these differential equations can be found in almost every other book on transmission lines and are therefore skipped here.

The impedance of the voltage and current waves is calculated as a characteristic impedance and the propagation delay through the circuit. It is the time delay (TD) of a sine wave entering the circuit until it comes out the other end.

The solution of the wave equation using sine waves results in:

$$Z_0 = \sqrt{\frac{L}{C}} = \sqrt{\frac{L_{\text{total}}}{C_{\text{total}}}} = \sqrt{\frac{L_{\text{Len}}}{C_{\text{Len}}}}$$

$$TD = \sqrt{L_{\text{total}} C_{\text{total}}} = n \times \sqrt{LC} = \text{Len} \times \sqrt{L_{\text{Len}} C_{\text{Len}}} \quad (2.12)$$

$$v = \frac{\text{Len}}{TD} = \frac{\text{Len}}{\text{Len} \times \sqrt{L_{\text{Len}} C_{\text{Len}}}} = \frac{1}{\sqrt{L_{\text{Len}} C_{\text{Len}}}}$$

These relationships are a direct result of solving the second-order linear differential equation.

It is from these methods that most engineers are taught the behavior of the characteristic impedance being the square root of L/C . But what does it mean?

While these equations result in the relationships between the characteristic impedance, time delay and speed of the signal, and the circuit elements of capacitance and inductance, it provides little physical insight into the behavior of signals on transmission lines.

What was presented in the previous section as a zen approach to analyzing the instantaneous impedance of a transmission line, with the distribution of small capacitor buckets, is NOT an equivalent electrical circuit model. It is a physics model.

We explicitly introduce the concept of the signal propagating at the speed of light in the material. We introduced the concept of small capacitor elements distributed some distance apart. Neither of these features are found in electrical circuit models.

From the physical model, we concluded the signal sees an instantaneous impedance that we used to define the characteristic impedance. We also calculated the time delay (TD) of the transmission line as:

$$TD = \frac{L_{len}}{v} \quad (2.13)$$

Using these two different models, we derived the same terms but from very different approaches. The values derived from each approach are exactly the same. They just include different elements.

In the physics model, we explicitly introduced the concept of the finite propagation speed of the signal, v . In the circuit model, we derived the speed of the signal in terms of the L_{Len} and the C_{Len} .

Using this connection, we can convert the physics model into the electrical model:

$$Z_0 = \frac{1}{C_{Len} \times v} = \frac{1}{C_{Len} \times \frac{1}{\sqrt{L_{Len} C_{Len}}}} = \sqrt{\frac{L_{Len}}{C_{Len}}} \quad (2.14)$$

Where did the inductance go in the physics model? It was hidden in the assumption about the speed of the signal. We explicitly included the speed as a property of the interconnect.

The characteristic impedance of the transmission line is both the square root of the ratio of the inductance to the capacitance in the

transmission line and it is the inverse of the capacitance per length times the speed of the signal. They are the same impedance written in slightly different forms.

While both approaches result in the same values for the important figures of merit of a transmission line, the physics approach offers some intuitive understanding of what environment the signal sees. The circuit model approach does not provide this same level of understanding, but has other applications, as is introduced in Chapter 8.

2.8 What is Special about 50 Ohms?

There is nothing fundamental or unique about 50 ohms. The only really fundamental impedance is the impedance of free space.

When a signal propagates in free space as light, it is a changing electric field and a changing magnetic field. It sees an impedance as it propagates.

The definition of impedance in circuits is based on

$$Z = \frac{V}{I} \quad (2.15)$$

This is really a circuit approximation of the fields. Voltage is an approximation for electric fields and current is an approximation for magnetic fields. The impedance a propagating electromagnetic field sees is related to the ratio of the electric to magnetic fields:

$$Z = \frac{E}{H} = \sqrt{\frac{\mu_0}{\epsilon_0}} \approx \sqrt{\frac{1.257 \times 10^{-6} \text{ N} \cdot \text{A}}{8.854 \times 10^{-12} \text{ F} \cdot \text{m}}} = 377 \, \Omega \quad (2.16)$$

We refer to this impedance as the characteristic *impedance of free space*. It is sometimes referred to as the characteristic impedance of the universe as it has fundamental significance in electrodynamics.

While it is important in electromagnetic applications, it has little importance when it comes to signals in the GHz range propagating on transmission lines built on circuit boards.

So why is 50 ohms so special and commonly used? The answer lies in the early use of coax cables in the communications and radar systems in the 1930s. This is described in [Microwave Tubes](#) by A.S. Gilmour. It is based on the geometry for lowest loss coax cables.

In the early days of radio, transmitters and receivers were not very efficient and every watt of power was precious. When transporting an RF signal from a TX to an antenna for broadcast, the goal was to use as low a loss cable as practical.

When a low-loss dielectric, such as polyethylene, with a $Dk = 2.3$, is used in a cable, the cable's losses are dominated by copper losses. The skin depth in copper is

$$\delta = \sqrt{\frac{\rho}{\pi f \mu_0}} \quad (2.17)$$

where

δ = the skin depth in m

ρ = bulk resistivity of copper, 1.7×10^{-8} ohm-m

f = the frequency in Hz

μ_0 = the permeability of free space = $4\pi \times 10^{-7}$ H/m

The series resistance of the cable, above about 1 MHz, when limited by skin depth, is the series resistance of the center conductor and outer shield. A larger diameter inner conductor means a lower resistance cable. The series resistance of the center and outer conductor, assuming the current is skin depth limited, is:

$$R_{Len} = \frac{\rho}{\pi\delta} \left(\frac{1}{d} + \frac{1}{D} \right) = \left(\frac{1}{d} + \frac{1}{D} \right) \sqrt{\frac{\rho f \mu_0}{\pi}} \quad (2.18)$$

where

R_{Len} = the resistance per length of the center conductor

d = the outer diameter of the center conductor

D = the diameter of the outer shield

ρ = the bulk resistivity of copper, 1.7×10^{-8} ohm-m

δ = the skin depth of copper

As the diameter of the center conductor increases, the resistance of the center conductor decreases, exactly as expected.

But the attenuation in a cable is about more than just the series resistance. The attenuation per length of a coax cable is:

$$\text{atten} [\text{dB} / \text{len}] = 4.34 \times \frac{R_{Len}}{Z_0} \quad (2.19)$$

where:

$\text{atten}[\text{dB}/\text{len}]$ = the attenuation of the signal in dB/length

R_{Len} = the resistance per length in ohms/length

Z_0 = the characteristic impedance of the cable.

The characteristic impedance of a coax cable is related to the dielectric material and the diameter of the inner conductor and inner dimension of the outer shield. For a coax cable, it is given by:

$$Z_0 = \frac{377 \Omega}{2\pi \sqrt{Dk}} \ln\left(\frac{D}{d}\right) \quad (2.20)$$

where

Z_0 = the characteristic impedance of the cable

Dk is the dielectric constant of the material inside the cable

D = the inner diameter of the shield

d = the outer diameter of the center conductor

For the lowest loss cable, we want the smallest series resistance and the largest characteristic impedance. This means as large an outer diameter as practical. If we fix the outer diameter to be the largest practical size, we can ask, is there an optimal center conductor for lowest loss?

The larger the center conductor, the lower the series resistance and the lower the loss. But, a larger center diameter with a fixed outer diameter shield conductor means a lower characteristic impedance, which means a higher attenuation.

Using these relationships, we can fix the outer shield to a diameter of 1 cm, for example, and vary the inner diameter of the conductor and calculate the resulting loss and the characteristic impedance. By plotting the attenuation against the characteristic impedance, we can find the optimum impedance that results in the lowest attenuation for the special case of polyethylene dielectric.

Figure 2.9 is a plot of the attenuation and characteristic impedance. The impedance for lowest attenuation using polyethylene is about 50 ohms.

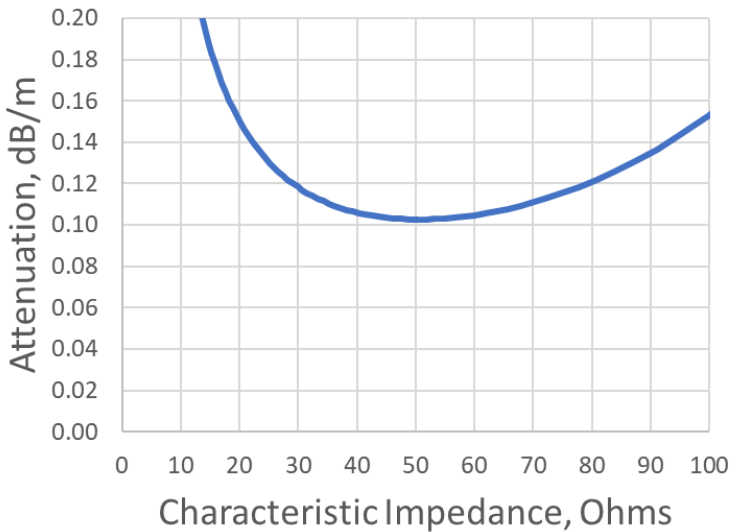


Figure 2.9 The attenuation versus the characteristic impedance of a coax cable with a fixed outer diameter with a polyethylene dielectric.

This is the origin of the use of 50 ohms as the characteristic impedance for cables.

Once the optimal characteristic impedance of a coax cable is fixed, the rest of the interconnect system, from the connectors, the receivers, the amplifiers, the drivers, and all the test equipment used to test all the components, should be the same 50 ohms impedance.

With the early industry moving toward 50-ohm components, it became more cost effective to engineer all systems to this 50 ohm value.

But 50 ohms is not necessarily the most cost-effective impedance environment for all electronic systems, especially wide bandwidth

high speed digital systems fabricated not as coax cables in polyethylene, but as planar interconnections in FR4 type polymers.

Some electronic products use a different impedance environment. Depending on which performance parameter is most important, a higher or lower characteristic impedance for the system may have an advantage. For example:

- A *higher impedance* means a higher termination resistance and lower power dissipation
- A *lower impedance* generally means lower cross talk
- Narrower traces and *higher impedance* can result in higher interconnect density
- *Lower impedance* lines can be a better match to the typically lower impedance of through-hole-vias and results in less reflection noise
- Wider lines and slightly *lower impedance* can sometimes result in lower attenuation

But, if there is no strong compelling reason otherwise, a 50-ohm target impedance for single-ended traces and 100-ohm differential impedance is a reasonable starting place for high-speed digital systems.

2.9 The Paradox of Current Flow into a Transmission Line

The signal is the voltage difference between adjacent points on the signal and return conductors. As it propagates down a transmission line, there is also a current associated with it. It's the combination of this voltage and current that creates the instantaneous impedance the signal sees.

If the characteristic impedance of a transmission line is 50 ohms and a 1V signal is launched into the transmission line, a current of $1\text{V}/50\text{ ohms} = 20\text{ mA}$ will be launched into the transmission line.

After the 20 mA current enters the signal conductor, when does the 20 mA current come out of the return path? This situation is illustrated in **Figure 2.10**.

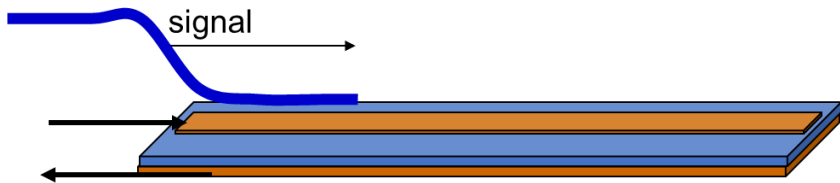


Figure 2.10 When the 1V signal is launched into the transmission line, and 20 mA is sent into the signal conductor when does the 20 mA come out the return conductor?

Keep in mind that there is an insulating dielectric between the signal and return conductor. Of course, free conduction charges cannot flow through the insulating dielectric; it's an insulating dielectric.

Based on the common principles taught in school about current flow, we would think that the 20 mA of current would enter the signal line and have to flow all the way down to the end of the transmission line before it could then connect to the return path and make its way back to the front of the transmission line and exit out of the return path.

But what if the far end of the transmission line is open? How does the current entering the transmission line at the front know that the far end is open and there is no connection to the return path? Would any return current come out the return conductor, ever?

The notions we learn early on of how current flows in a circuit are only suitable for what happens at DC. They do not guide our understanding in the high-speed, time domain world.

2.10 Displacement Current as a New Type of Current

To understand the behavior of currents in a transmission line, we have to introduce a new type of current, displacement current. This is the basis of how current flows through a capacitor.

One of the many contributions James Clerk Maxwell made to unifying the concepts of electromagnetics is realizing there are two types of currents. There is current composed of free conduction charges, which is what we usually think of when we think current, and there is a current that is due to a changing electric field, which he called *displacement current*.

This new current was needed to provide continuity to the current flow through a capacitor. Of course, at DC, there is no current flow through a capacitor, since there is an insulating dielectric between the two conductors that make up the capacitor.

The definition of the capacitance of a capacitor is that:

$$C = \frac{Q}{V} \quad \text{or} \quad Q = CV \quad (2.21)$$

Capacitance is a measure of the efficiency for two conductors to store charge at the cost of voltage. A large capacitance means the conductor geometry is very efficient at storing a lot of charge for a small voltage.

To increase the voltage across a capacitor, we must add some more excess positive charge to the top conductor and more negative charge to the bottom conductor.

When negative charge is added to the bottom conductor, this is the same as pushing positive charge out of the bottom conductor.

When the voltage across a capacitor is increased, it looks from the outside like positive charges flow into the top conductor and positive charges flow out of the bottom conductor. This is illustrated in **Figure 2.11**.

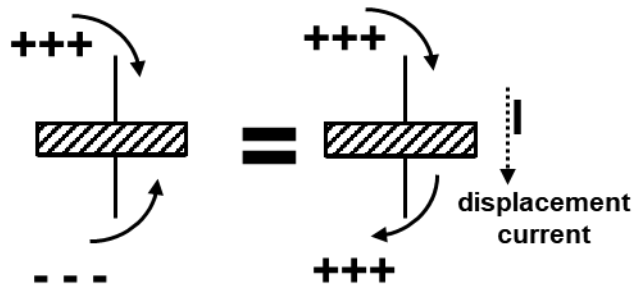


Figure 2.11 When the voltage across a capacitor is increased, positive charge goes into the top conductor and positive charge flows out of the bottom conductor. Displacement current keeps the current continuous in the circuit.

Do actual free charges move through the insulating dielectric between the top and bottom conductors? Of course not. There is an insulating dielectric between the conductors. But it sure looks that way.

To James Clerk Maxwell, when the voltage across a capacitor changed, it looked like current flowed into and out of the terminals of the capacitor. To maintain continuity of current, which was a cornerstone of his unified theory, he required a current to flow through the insulating dielectric.

In his vision, he saw all dielectric insulating materials composed of a sea of positive and negative charges, bound together by local electric forces. When an external electric field was applied, these charges were pulled slightly apart. He termed this behavior, *polarization*.

When the external electric field polarized the material, the charges were slightly displaced. This motion of the positive and negative charges when their separation changes is a current. Positive charges moving one way and negative charges the opposite is a net, positive current.

When the external electric field changes, the polarization changes and the movement of the bound charges, the displacement of the charges, is a current. When the electric field stops changing, the material is still polarized, but the bound charges are not moving.

He called this current from the changing polarization, *displacement* current.

It was easy to understand for a dielectric material which could be polarized. But how could this apply to a vacuum in which there is no material to polarize?

In his 1861 perspective of the universe, all space, even a vacuum, was filled with a medium referred to as the ether. It was the medium which permeated everywhere and through which light traveled. It was thought to be composed of small charged particles bound together, but polarizable, just like a dielectric material.

When an electric field is applied to this ether-dielectric, its positive and negative charges, still bound to each other, are polarized. When the electric field changes, displacement current flows through the ether as the bound charges' separation changes.

In Maxwell's view, a changing electric field was a displacement current just as real as a conduction current. Displacement current from changing electric fields affects magnetic fields just as much as conduction current. Displacement current provides the continuity of current with conduction current.

The displacement current density in a material is exactly equivalent to the changing electric field in the material:

$$J_d = \frac{\partial D}{\partial t} = \epsilon_0 \frac{\partial E}{\partial t} + \frac{\partial P}{\partial t} \quad (2.22)$$

where

J_d = the displacement current density

D = the electric displacement field in the material

ϵ_0 = the permeability of free space

E = the electric field

P = the polarization of the material

The first term is the displacement current from the changing polarization of free space, the ether in Maxwell's view. The second term is the displacement current from the changing polarization of the dielectric of the material, related to the dielectric constant of the material.

When you see the electric field lines that connect two conductors at different voltages, and the voltage between them increases, displacement current will literally flow along the electric field lines that will change. This is illustrated in **Figure 2.12**.

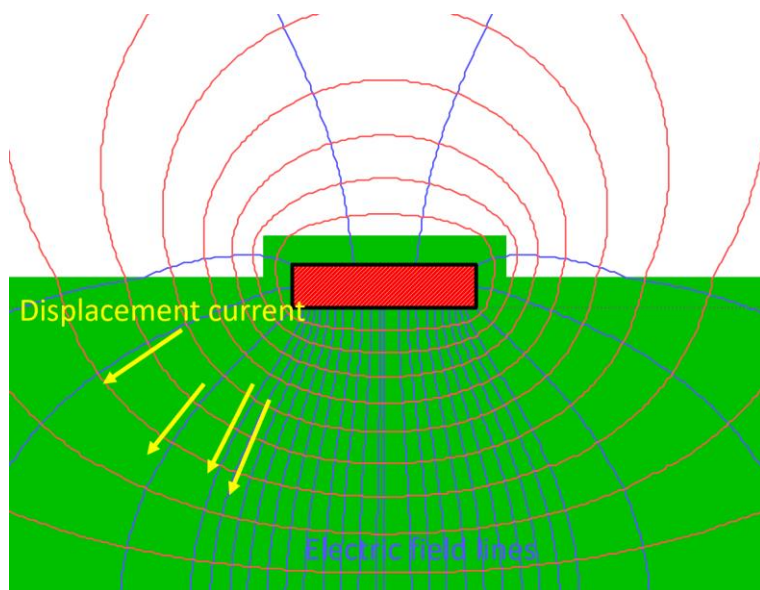


Figure 2.12 Displacement current flows along the electric field lines when the voltage between the conductors change.

The higher the electric field line density, the higher the displacement current density when the electric field changes. The faster the voltage changes, the larger the electric fields change with time and the larger the displacement current density.

In a capacitor, if 20 mA of current flows into the top conductor and 20 mA of current flows out the bottom conductor, 20 mA of displacement current will flow through the insulating dielectric.

As the applied voltage to a capacitor increases, the electric field inside the dielectric between the conductors increases and the displacement current flows.

This is precisely how the return current returns between the signal and the return conductor in a transmission line.

2.11 Return Current in a Transmission Line

When a voltage is launched in a transmission line, as the voltage at the beginning of the line increases, the electric field between the signal and return conductors increases and displacement current flows to connect the input current into the signal path to then instantly return out the return path. This current loop is shown in **Figure 2.13** at two instances in time.

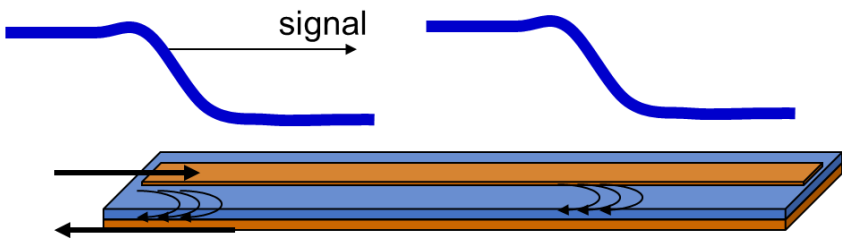


Figure 2.13 At any instant in time, displacement current flows between the signal and return path **ONLY** where the rise edge of the signal is.

As the wavefront moves down the transmission line, the only place that displacement current flows is where the changing voltage wavefront is located. As this voltage wavefront moves down the transmission line, a coincident displacement current wavefront also moves down the transmission line.

When 20 mA of current flows into the signal path, when does 20 mA of current flow out of the return conductor? Instantaneously. It flows between the signal conductor and the return conductor through the displacement current of the changing electric field that is created by the changing voltage.

This current wavefront has two directions associated with it. It has a direction of propagation, which is the same as the direction of propagation of the signal, and it has a direction of circulation.

When the signal is a positive edge, propagating from the left to the right, the current wavefront circulates in the clockwise direction, advancing as signal-return, signal-return, signal-return.

If the voltage were a negative-going voltage, also propagating from the left to the right, the direction of circulation of the displacement current would be opposite, from the return conductor to the signal conductor, circulating in the counterclockwise direction.

The observation that the current flow in a transmission line has two directions, and they are independent of each other, is a critically important principle. Understanding this behavior will resolve what may otherwise appear as paradoxes in transmission line behavior.

Watch this video and you can see the dynamic nature of the signal-return current wavefront propagating down a transmission line.

2.12 Where Does the Return Current Flow?

If the return conductor is a wide plane, like a ground plane, the actual path the return current takes will not spread out everywhere in the plane.

Above about 100 kHz, all the return current will flow directly underneath the signal current. The path the current takes is driven by the path that reduces the impedance the current loop sees.

At DC, the impedance the current sees is all about the DC resistance of the signal and return paths. The current will spread out in the return plane at DC finding the lowest resistance path.

But at any other frequency, the impedance the current sees is the series combination of the series resistance and the loop inductance of the signal-return path loop:

$$Z = R + j\omega L \quad (2.23)$$

Above about 100 kHz, the inductive impedance is higher than the resistive impedance and the current will redistribute to take the path of lowest loop inductance.

The lowest loop inductance translates to:

- As far apart for currents in the same direction
- As close together for currents traveling in the opposite direction.

This is literally the same driving forces giving rise to currents flowing in the outer surface, or skin of the conductor, referred to as the *skin effect*.

It is difficult to calculate the current distribution for microstrip or other planar geometries. This requires a 2-D full-wave field solver. **Figure 2.14** shows an example of the current distribution in the same microstrip transmission line at three different frequencies. As the frequency increases, the current in the return path flows mostly under the signal line. It does not spread out.

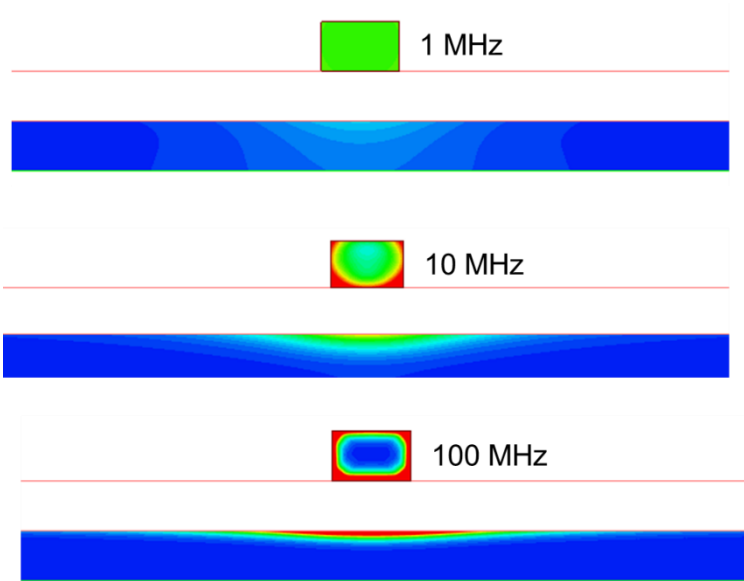


Figure 2.14 The calculated current density in a transmission line at three different frequencies. Dark blue means no current and red means very high current density.

This means that as the signal traces on a board route around the board, the return current distribution in the return plane will also route around and follow the signal path, staying directly underneath. This is illustrated in another simulation of the current distribution in a microstrip that makes a sharp bend in its route shown in **Figure 2.15**.

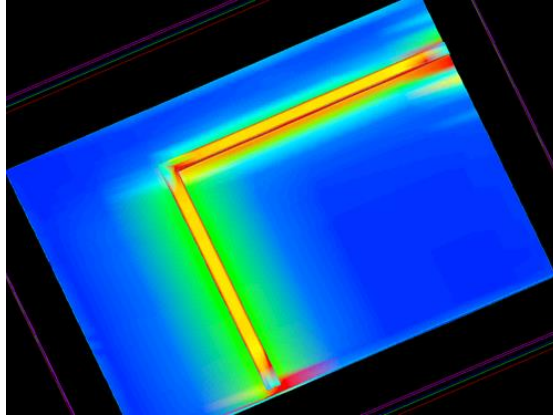


Figure 2.15 Current distribution in the return plane when a signal traces makes a sharp turn. Note the return current stays directly beneath the signal conductor.

2.13 Review Questions

1. Why is it a good idea to not call the return path ground?
2. What is generated between the signal and return conductors at the beginning of a transmission line when a signal is launched? What will happen?
3. Is the signal the moving electrons in the conductor or the electric field in the dielectric?
4. What is a good rule of thumb for the speed of a signal in an FR4 transmission line?
5. What is the difference between the voltage measured by an oscilloscope and the signal on the transmission line?
6. Using the wiring delay of a transmission line, what is the time delay of a 1-inch long interconnect in a large BGA package? A 3-inch-long interconnect on a small circuit board? A 30-inch-long interconnect in a backplane?

7. What does instantaneous impedance refer to? Why is this so important?
8. If an interconnect trace gets wider, what happens to the capacitance per length? What happens to the instantaneous impedance?
9. What is required by an interconnect to have a characteristic impedance?
10. In the physics model of a transmission line, where did the inductance of the interconnect go?
11. What is special about 50 ohms and why is it so common in high-speed digital applications?
12. When 20 mA of current flows into a signal conductor that is 12 inches long, when will the return current come out of the return conductor?
13. What is displacement current and where does it flow?
14. Where in the return path will the return current flow?